Session 1.4

Workhorse Macroeconomic Models

The models this course solves, the object each one asks for, and where grids stop working



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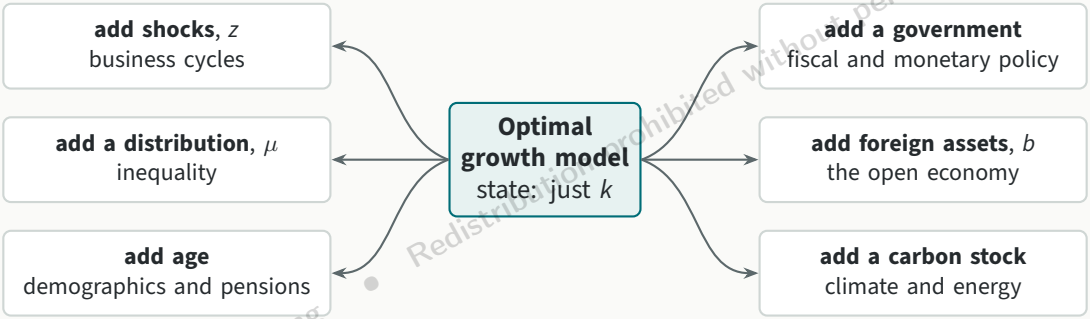
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Why This Deck, and Who It Is For

- The course is built for macroeconomists. The methods carry over to dynamic games in industrial organisation, asset pricing, and labour search.
- What AI will do for us: **(1)** solve the workhorse models efficiently and at scale; **(2)** turn unstructured data into measurement; **(3)** help us understand and write models, design algorithms, and implement code.
- **(4)** Verification, judgment and responsibility stay with the researcher.
- A word of warning: in this deck an **agent** is a household or a firm inside a model. That is the word's third meaning today, after the AI agent and the agent of contract theory.
- **For each model, ask: what is the state, what object must be computed, and what check would expose a wrong answer?**

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One Model, Many Directions



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- The trunk is one planner choosing consumption and capital. Each branch is a field of macroeconomics.
- **Every branch adds state variables, and more state means harder to compute.**

The Environment

$$\max_{\{c_t,\, k_{t+1}\}} \sum_{t=0}^{\infty} \beta^t \log c_t \qquad \text{subject to} \qquad c_t + k_{t+1} = A k_t^{\alpha}, \qquad k_0 \text{ given.}$$

- One good, one decision-maker. Output Ak^{α} is either eaten (c) or saved as tomorrow's capital (k').
- A is **total factor productivity** (TFP), α the capital share, β the discount factor.
- Log utility and full depreciation are chosen on purpose: this version has a closed-form solution.
- **State**: what you must know today to decide (k). **Control**: what you choose (k').
- **One calibration wherever this model returns (today's lab, Session 2): $\alpha = 0.33$, $\beta = 0.96$, $A = 1$.**

Source: Brock & Mirman, "Optimal economic growth and uncertainty: the discounted case", *Journal of Economic Theory* (1972).

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Why the Bellman Equation Can Be Trusted

$$V(k) \;=\; \max_{0<k'<Ak^{\alpha}} \Big\{ \log\!\big(Ak^{\alpha} - k'\big) + \beta\, V(k') \Big\} \;\equiv\; (TV)(k)$$

Result	What it guarantees
Principle of optimality	the sequence problem and the Bellman equation share a solution
Blackwell's conditions	monotonicity and discounting make T a contraction, modulus β
Contraction mapping	one fixed point V; $V_{n+1} = TV_n$ reaches it from any V_0
Theorem of the maximum	V is continuous, and an optimal choice exists at every k
Concavity	V strictly concave, so the policy $k' = g(k)$ is a function
Benveniste–Scheinkman	V differentiable; the envelope condition gives the Euler equation

- **Two descriptions of one solution, a fixed point of T and the Euler equation: two ways to compute it.**

Source: Bellman (1957); Blackwell (1965); Benveniste & Scheinkman, *Econometrica* (1979); Stokey, Lucas & Prescott (1989), ch. 4.

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Solving It, Approach 1: Iterate on the Value Function

Value function iteration

1. Grid $k_1 < \cdots < k_N$; guess $V_0 \equiv 0$; tolerance ε.
2. For every k_i, the Bellman update:

$$V_{n+1}(k_i) = \max_{k_j < A k_i^{\alpha}} \left\{ \log(A k_i^{\alpha} - k_j) + \beta V_n(k_j) \right\},$$

   and record the maximiser $g_{n+1}(k_i) = k_j$.
3. If $\max_i |V_{n+1}(k_i) - V_n(k_i)| < \varepsilon$, stop; otherwise $n \leftarrow n + 1$, go to 2.
4. Return V_{n+1} and g_{n+1}.

- Convergence is guaranteed: the distance to V shrinks by β each step; from $V_0 = 0$ it takes 417 steps to reach 10^{-6} (step 3's rule stops after about 340).
- The policy settles long before the value. **Policy iteration** (Howard 1960) exploits this.
- **Cost per step: $N \times N$ comparisons. Session 3 replaces the grid with a network.**

Source: Bellman (1957); Howard (1960). Iteration count at $\alpha = 0.33$, $\beta = 0.96$, $A = 1$.

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Solving It, Approach 2: Solve the Equilibrium Conditions

Time iteration on the Euler equation

1. Grid $k_1 < \cdots < k_N$; guess a policy g_0, e.g. $g_0(k) = \frac{1}{2} A k^{\alpha}$.
2. For every k_i, solve for k':

$$\frac{1}{A k_i^{\alpha} - k'} = \beta \frac{\alpha A k'^{\alpha - 1}}{A k'^{\alpha} - g_n(k')},$$

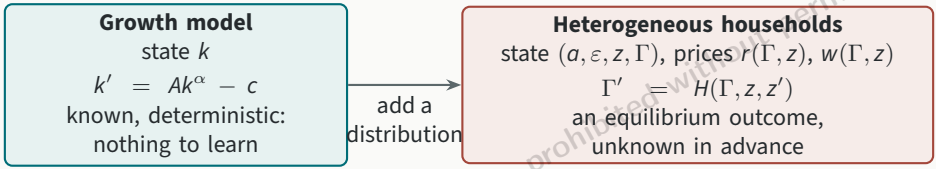
   with g_n interpolated; set $g_{n+1}(k_i) = k'$.
3. Stop once $\max_i |g_{n+1}(k_i) - g_n(k_i)| < \varepsilon$.
4. Check the Euler residual on fresh k off the grid.

Source: Coleman (1990); Judd, *JET* (1992); Carroll, *Economics Letters* (2006).

- The unknown is the policy g, never V.
- Variants: projection (Judd 1992) sets the residual to zero at nodes; endogenous grids (Carroll 2006) invert the equation, with no root-finding.
- **Euler alone admits explosive paths; transversality rules them out. Session 2 trains a network on the squared residual.**

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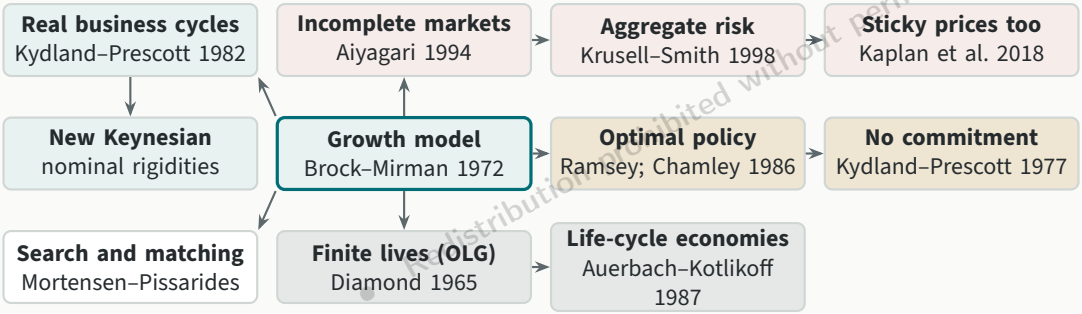
When the Environment Is Unknown



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- In reinforcement-learning terms the **environment** is the transition law and the reward.
- Γ is the distribution of households over wealth and productivity. Its law of motion H comes from everyone's choices, so it cannot be written down before the household problem is solved.
- Classical answer: guess H from a few moments (Krusell–Smith, later in this deck).
- Reinforcement learning: learn from simulated transitions (s, a, r, s'). H is known to the simulator, never to the learner, and expectations become sample averages.
- **Trivial in the growth model; the central difficulty once a distribution is a state.**

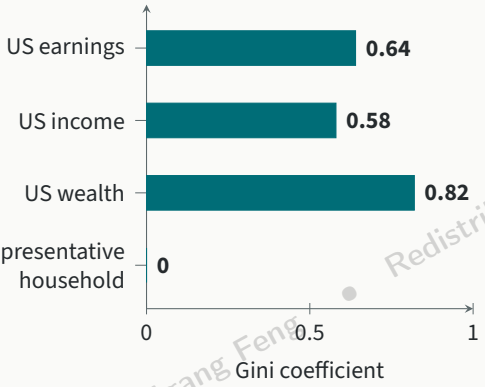
A Family Tree of Workhorse Models



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- Teal: one representative household. Red: heterogeneity. Gold: policy. Grey: the life cycle.
- **Every branch now has a deep-learning frontier.** Sessions 2 to 4 walk the red, gold and grey ones.

A Puzzle: Wealth Is Far More Unequal Than Income

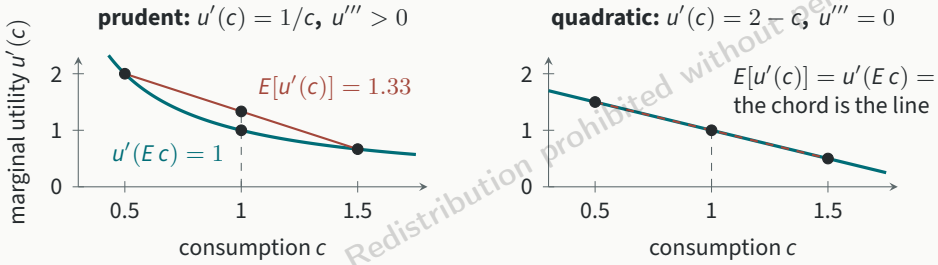


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- **Gini coefficient**: 0 if everyone holds the same, 1 if one household holds everything.
- The top 1% hold about a third of US wealth. A representative household has no distribution at all.
- The textbook incomplete-markets model gets about half of the wealth Gini or less: 0.32–0.41 against 0.82.
- **Every candidate fix (entrepreneurs, bequests, unequal returns) is a new state variable.**

Source: Survey of Consumer Finances 2007, in Díaz-Giménez, Glover & Ríos-Rull, *FRB Minneapolis Quarterly Review* (2011).

Why Households Self-Insure: Prudence, Not Risk Aversion



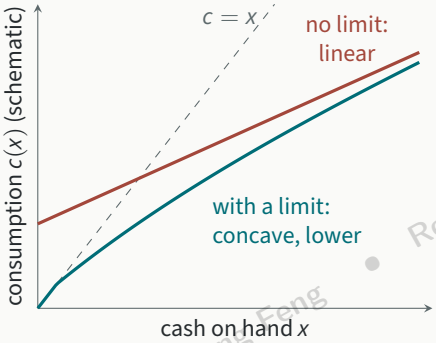
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- Both are risk averse ($u'' < 0$); c_{t+1} is 0.5 or 1.5 with equal odds. In $u'(c_t) = \beta(1+r)E[u'(c_{t+1})]$, risk raises the right side only on the left.
- Left: saving rises. Right: saving is as under certainty (**certainty equivalence**).
- **Precautionary saving needs prudence ($u''' > 0$), not risk aversion alone.**

Source: Leland (1968); Sandmo (1970); Kimball, *Econometrica* (1990).

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A Second Motive: The Borrowing Limit



- Quadratic utility, $u''' = 0$. Without a limit, $c(x)$ is linear and risk does not move it.
- With a limit, a bad draw tomorrow cannot be smoothed by borrowing. The household consumes less today, most of all when its cash on hand is low.
- **In Aiyagari both motives operate: prudence and the borrowing limit.**

Source: Zeldes, *Quarterly Journal of Economics* (1989); Carroll & Kimball (2001). Curves are schematic.

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Aiyagari: The Household and the Firm

$$v(a,\varepsilon) = \max_{c,\,a'} \left\{ u(c) + \beta \sum_{\varepsilon'} \pi(\varepsilon' \mid \varepsilon)\, v(a',\varepsilon') \right\} \quad \text{s.t.} \quad c + a' = (1+r)a + w\varepsilon, \qquad a' \geq -\underline{a}.$$

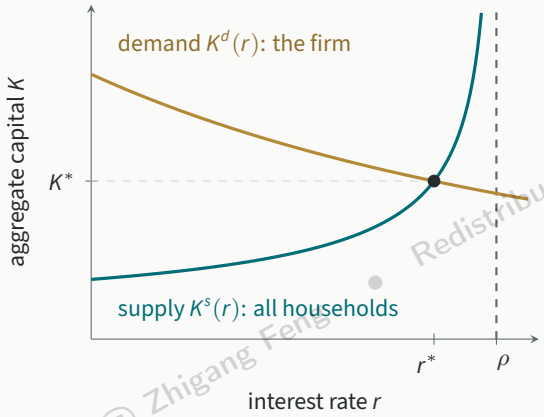
$$r = \alpha A\,(K/L)^{\alpha-1} - \delta, \qquad w = (1-\alpha) A\,(K/L)^{\alpha}.$$

- a: assets. ε: labour productivity, a Markov chain no one can insure. $\underline{a}$: the borrowing limit.
- A continuum of households, identical ex ante, different ex post: their histories differ.
- Each household's problem is a **Markov decision process**: state (a, ε), action a', reward $u(c)$, discount β, transition π.
- **The new object is the cross-section: a stationary distribution $\lambda(a, \varepsilon)$ of households.**

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Source: Aiyagari, "Uninsured idiosyncratic risk and aggregate saving", *Quarterly Journal of Economics* (1994).

Equilibrium Is Where the Curves Cross

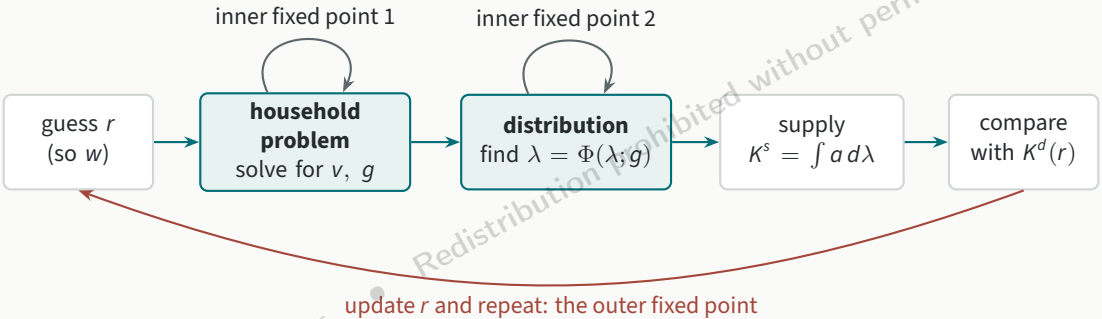


- $\rho = 1/\beta - 1$ is the rate of time preference. With full insurance, $r = \rho$.
- Savers facing risk would accumulate without bound as $r \to \rho$, so supply is finite only below it.
- Hence $r^* < \rho$, and capital sits above its complete-markets level.
- **That extra capital is not a free lunch. It is the cost of self-insurance.**

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Source: Aiyagari (1994). A schematic: demand from the firm's first-order condition, supply asymptotic to $\rho = 1/\beta - 1$.

The Nested Fixed Point



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- Every point on the supply curve contains two solved problems: a Bellman equation and an invariant distribution. Add a second price, or a tax, and the outer search becomes a system.
- **A small error at the outer level can hide a failure inside either inner problem.**

Krusell–Smith: The Distribution Starts to Move

Stationary world (Aiyagari). The distribution λ^* is an unknown we solve for *once*. Prices are numbers. Households forecast nothing.

add a TFP shock z

With aggregate shocks. The distribution Γ_t is a *state that moves every period*, $\Gamma' = H(\Gamma, z, z')$. Prices move with it. Households must forecast a distribution.

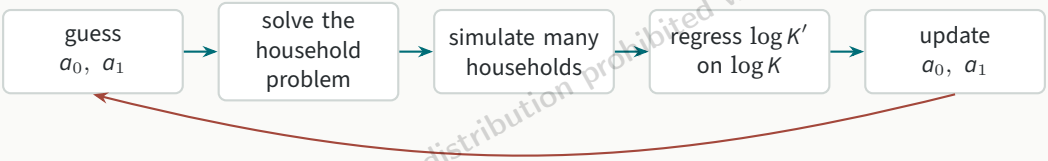
- Γ: the cross-sectional distribution of households over assets and productivity.
- Put wealth on 100 grid points and employment on 2. Γ is then 200 numbers that sum to one: a point in a 199-dimensional simplex.
- In Aiyagari we solved for one such point. Here we need a *function* of it.
- **The state is no longer a vector. It is a distribution.**

Source: Krusell & Smith, "Income and wealth heterogeneity in the macroeconomy", *Journal of Political Economy* (1998).

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The Krusell–Smith Bet

$$\log K' \;=\; a_0(z) + a_1(z)\,\log K$$

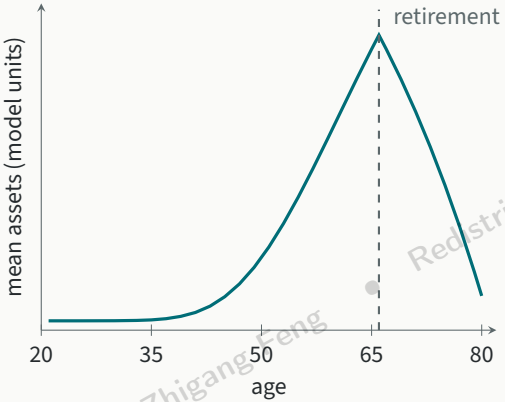


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- The bet: households forecast with one moment, mean capital K, not all of Γ. The fit is $R^2 > 0.9999$: **approximate aggregation**.
- R^2 **can flatter.** It scores one-step forecasts; errors that build up over many periods hide behind it (Den Haan 2010).

Source: Krusell & Smith (1998); Den Haan, *Journal of Economic Dynamics and Control* (2010).

OLG: Age Becomes a State

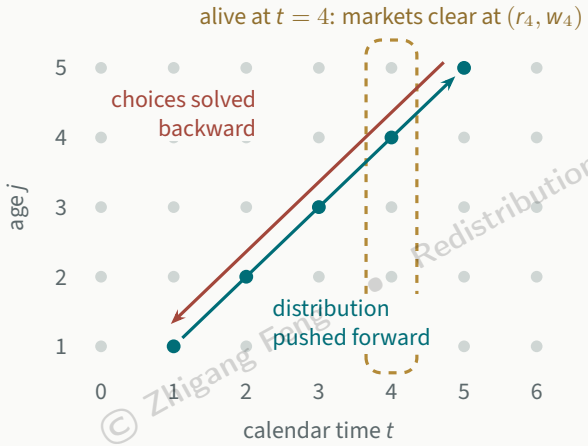


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- **Overlapping generations** (OLG): people live finite lives, and cohorts of different ages coexist.
- Sixty cohorts, ages 21 to 80, work to 65: they save for retirement, then run assets down.
- Finite lives help: solve backward from the last age, with no inner fixed point in values.
- **The state now carries age, so pensions, taxes and demographics can be studied.**

Source: After Auerbach & Kotlikoff (1987; 55 cohorts, to age 75). Profile: the instructor's 60-cohort model with a flat pension.

Transitions: Households Backward, Distributions Forward



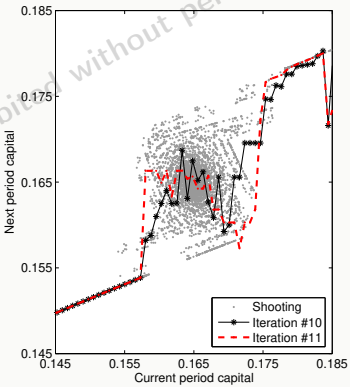
- A reform changes the whole price path $\{r_t, w_t\}$, not one number.
- Guess the path. Solve each cohort **backward** from its last age.
- Push the distribution **forward** from today's cross-section.
- Check market clearing in *every* period, update the path, repeat.
- **The unknown is a sequence of prices and distributions.**

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Source: Auerbach & Kotlikoff (1987); Conesa & Krueger, "Social security reform with heterogeneous agents", *RED* (1999).

The Government Is a Player Too

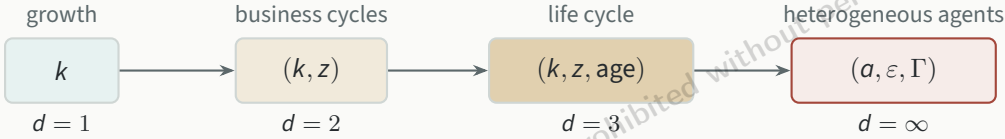
- **Ramsey policy**: choose the whole tax path at date 0. Households' Euler equations become constraints.
- **Commitment adds a state**: a past promise, with a feasible range unknown in advance.
- **Time inconsistency** (Kydland–Prescott 1977): tomorrow's government will not keep today's plan.
- **Theory warns us.** A continuous equilibrium can fail to exist. A method that assumes a smooth policy then returns a confident, wrong answer.



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Source: Santos, *Journal of Economic Theory* (2002). Right: his example computed numerically; iterations do not settle.

What Are We Solving For? The Objects Escalate



Model	Object to compute	Computational problem
Growth	value $V(k)$ and policy $g(k)$	one fixed point
Aiyagari	policies, a price r, a distribution λ	a nested fixed point
Krusell–Smith	policies and a law $\Gamma' = H(\Gamma, z, z')$	a fixed point in forecasting rules
OLG, Ramsey	policies by age, price paths, tax rules	backward and forward; a fixed point in rules

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Why Grids Break

1 state: 10^2 points



2 states: 10^4 points



5 states: 10^{10} points



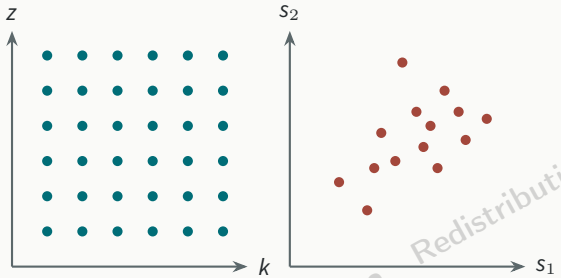
states d	grid points	one value table
2	10^4	0.08 MB
4	10^8	0.8 GB
6	10^{12}	8 TB

- **Curse one, the grid:** 100 points per state gives 100^d points.
- **Curse two, the expectation:** with 7 nodes per shock, 3 shocks cost $7^3 = 343$ terms at every grid point, every iteration.
- **A distribution as a state makes d effectively infinite.**

Source: Bellman (1957) coined "the curse of dimensionality". Dots are a schematic sample, not the literal count.

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Where AI Enters (1): From Grids to Samples



classical
a value at every grid point

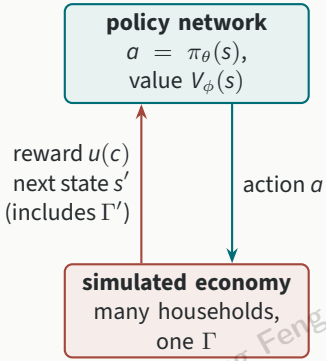
learned
a network fit on visited states

- A neural network stands in for the policy or value function.
- All equilibrium conditions enter one loss, minimised by gradient descent.
- Training states come from simulating the model: **you pay only for the states the economy visits.**
- So Γ need not be collapsed to a few moments.

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Source: Maliar, Maliar & Winant, *JME* (2021); Azinovic, Gaegauf & Scheidegger, *IER* (2022); Han, Yang & E, *QE* (2026); Fernández-Villaverde, Nuño & Perla, NBER WP 33117 (2024).

Where AI Enters (1), Continued: Reinforcement Learning



- **Critic**, target $\bar{\phi}$ held fixed:

$$\min_{\phi} \mathbb{E}\left[\left(u(c) + \beta V_{\bar{\phi}}(s') - V_\phi(s)\right)^2\right]$$

- **Actor**, along simulated paths:

$$\max_{\theta} \mathbb{E}\left[u(c_\theta) + \beta V_\phi(s'_\theta)\right]$$

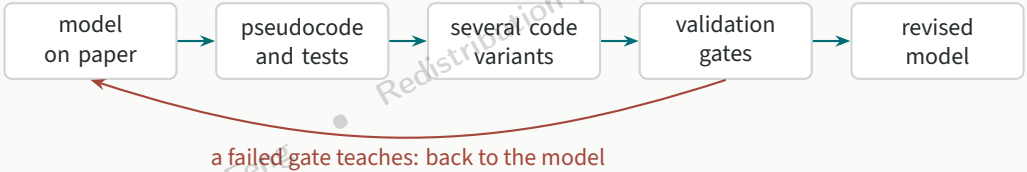
- No transition law is written down: expectations are simulated averages.
- **Equilibrium: every household uses the trained rule. Sessions 3 and 4 build it.**

Source: Sutton & Barto, *Reinforcement Learning* (2018), ch. 13; Miao & Feng, *AI for Economists*, ch. 11.

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Where AI Enters (2) and (3): Data, and a Collaborator

- **(2) Unstructured data become measurement.** Central-bank text gives a policy-stance index, night lights proxy for output, newspapers yield an uncertainty index.
- **(3) AI as a collaborator** helps you understand and write models, design algorithms, and implement code. Session 6 develops this loop.

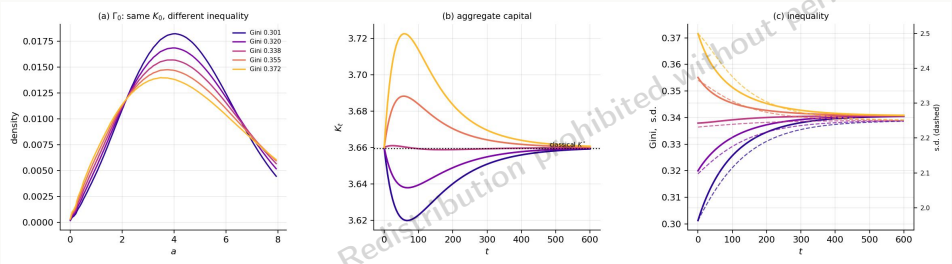


- **AI writes variants quickly. You choose the state, the equilibrium concept, the benchmark, and what a failure means.**

Source: Hansen, McMahon & Prat (2018); Henderson, Storeygard & Weil (2012); Baker, Bloom & Davis (2016).

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Research in Practice: Every Start Ends in the Same Place



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- Five Aiyagari economies: the *same* $K_0 = 3.660$, Gini 0.301 to 0.372. All reach the classical $K^* = 3.6595$.
- A network trained on $\Gamma_{t+1} = \Phi(\Gamma_t)$ makes convergence a result, not an assumption.
- **Same K_0, yet the distributions move for hundreds of periods. K alone misses it.**

Source: Feng, Han, Sargent & Zhu (2026), work in progress. Terminal K agrees to 3.5×10^{-8}.

What Stays With the Researcher

- A converged training loss certifies nothing. A solution is certified by a ladder of checks:
  - **Feasibility**: every choice respects the budget and borrowing constraints.
  - **Residuals on fresh draws**: unit-free Euler and Bellman errors, on states not used in training.
  - **A classical benchmark**: value function iteration on the low-dimensional case, where it still works.
  - **Theory**: policies are monotone, values concave, consumption smooth.
- The top rung is not computational: **your name on the result**.
- Today's lab climbs the first three rungs on the growth model, in under a minute.
- **Verification, judgment and responsibility stay with the researcher.**

Source: Feng, "AI for Economics and the Economics of AI", working paper (2026), §3.3; Judd, Maliar & Maliar (2017).

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Where Each Model Returns

Session	Object	What changes
1	Orientation	AI as tool, as object, and as an agent with incentives
2	Deep learning	Functions become trainable; equations become losses
3	Reinforcement learning	Sequential decisions are learned from interaction
4	Heterogeneous agents	Distributions and equilibrium become state variables
5	Text and LLMs	Language becomes data, measurement, and generation
6	RAG and agents	Retrieval, tools, traces, and governance in one workflow

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- The growth model returns in Sessions 2 and 3; Aiyagari, Krusell–Smith, OLG and optimal taxation in Session 4.
- **Next: a first lab that solves the growth model with a network, and checks it. Deck 1.5 sets up the workbench it runs on.**

Lab: Two Machines, One Environment



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- Machine A minimises the squared Euler residual of Approach 2.
- Machine B learned the *look* of FOMC statements, not facts: no residual to compute.
- **Same technology, two epistemic positions: one output you can verify, one you cannot.**